Thermodynamic Operating Constraints of Mammalian Somatic Cell Classes

A First-Principles Derivation from DNA Methylation Maintenance Information Costs

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Abstract

We derive thermodynamic lower bounds on DNA methylation entropy for eight mammalian somatic cell architecture classes — pluripotent stem, adult tissue stem, progenitor/transit-amplifying, terminally differentiated, rapidly cycling epithelial, immune effector, secretory/glandular, and stromal — from the Landauer cost of irreversible information maintenance at physiological temperature. The global information maintenance floor, $E_{\text{floor}} = N_{\text{CpG}} \cdot k_{\text{B}} T_{\text{body}} \ln 2 = 5.82 \times 10^{-14}$ J per cell division, sets a minimum Shannon entropy $H_{\min}$ below which no cell class can sustain epigenomic identity. We formalize a dimensionless epigenomic fidelity index $\mathcal{A} = H_{\text{act}}/H_{\min}$ that normalizes observed methylation entropy against the class-specific floor. Using Markov chain Monte Carlo (MCMC) with the *emcee* ensemble sampler (5 independent chains, R-hat < 1.01 for all 8 parameters, 8×10^5 posterior samples), we validate $H_{\min}$ against 37 published reference cell measurements spanning the Roadmap Epigenomics Consortium, ENCODE, and primary literature datasets. Posteriors are internally self-consistent across 6 of 8 classes within 1σ; the immune class correction of $0.795 \rightarrow 0.839$ is confirmed at 6.44σ, resolved by inclusion of neutrophil reference data. In a zero-free-parameter forward prediction, the $\mathcal{A}$ framework correctly identifies the direction of methylation entropy departure in 27 of 28 matched tumour–normal pairs (n = 4,304 matched samples) from the TCGA Pan-Cancer Atlas. Terminal-class cancers (glioblastoma, lower-grade glioma) show the largest thermodynamic departures ($\Delta\mathcal{A} = 0.228$–$0.273$), consistent with the prediction that the most epigenomically constrained cell class produces the largest entropy gap upon malignant transformation. We additionally fit the $E(a_{\text{bio}}) = \exp(1 - 1/a_{\text{bio}})$ activation function to published DunedinPACE age-stratified data, recovering a biological actualization ceiling $t_{\max}$ consistent with the Gompertz–Makeham human lifespan limit. Extended validation against published methylation data from Alzheimer's disease and Type 2 diabetes demonstrates that the $\mathcal{A}$ framework places each condition in its correct thermodynamic tier relative to the architecture floor, without requiring cancer training data. These results constitute the first first-principles thermodynamic specification of mammalian somatic cell architecture classes, providing a physically grounded reference frame for interpreting methylation data in cancer detection, neurodegeneration, aging research, pharmacology, regenerative medicine, and developmental biology.

Keywords: DNA methylation; Shannon entropy; Landauer principle; epigenomic fidelity; cell architecture; thermodynamics of information; MCMC; cancer detection; Alzheimer's disease; biological aging

Data availability: The G-002, G-008, n_{bio} ordering, and $E(a_{\text{bio}})$ MCMC validation scripts, together with the full evidence database (SQLite and JSON) and provenance hashes, are available at `https://github.com/hmahaffeyges/IAM-Validation` (Zenodo DOI: 10.5281/zenodo.19547624). Additional validation reports and cross-domain applications of the information actualization framework are available at `https://iamperformance.net`. Requests for access to the analytical engine should be directed to `hmahaffeyges@gmail.com`. All primary beta values are drawn from publicly available datasets cited in the text.

Contents

1 Introduction 5

2 Theoretical Framework 6

2.1 Landauer cost of DNA methylation maintenance . . . 6

2.2 Methylation entropy and the architecture floor . . . 7

2.3 The global floor and the commitment spectrum . . . 7

2.4 Metabolic sensitivity parameter . . . 8

2.5 Three-component decomposition . . . 8

2.6 Biological activation function . . . 9

3 Methods 9

3.1 Reference database construction . . . 9

3.2 MCMC validation of H_{min} values (G-002) . . . 10

3.3 Cancer floor breach prediction (G-008) . . . 10

3.4 Metabolic sensitivity ordering test (n_{bio} validation) . . . 10

3.5 DunedinPACE activation function fit . . . 11

3.6 Reproducibility . . . 11

4 Results 11

4.1 Thermodynamic specification: the terminal class as exemplar . . . 11

4.2 Extended validation: neurodegeneration and metabolic disease . . . 11

4.2.1 Alzheimer's disease — terminal class departure without malignant transformation . . . 13

4.2.2 Type 2 diabetes — secretory class failure . . . 14

4.2.3 DCIS stratification — pre-invasive breast cancer . . . 14

4.3 H_{min} posterior values and MCMC convergence (G-002) . . . 15

4.4 Architecture class specification . . . 16

4.4.1 Terminal class: the most constrained architecture . . . 17

4.4.2 Pluripotent class: high entropy by design . . . 17

4.5 Zero-free-parameter cancer validation (G-008) . . . 17

4.5.1 Three structural confirmations from neural biology . . . 17

4.6 Metabolic sensitivity ordering validation . . . 19

4.7 DunedinPACE activation function fit . . . 19

5 Discussion 20

5.1 What this framework provides that statistical approaches cannot 20
5.2 Applications 20
5.3 Limitations and experimental validation paths 22
5.4 Relationship to existing information-theoretic biology 22

6 Patent protection and data availability 23

7 Conclusion 23

S1 G-002 Reference Cell Database 31

S2 MCMC Chain Specifications 33

S3 Sensitivity analysis: H_{min} uncertainty 34

1. Introduction

DNA methylation at CpG dinucleotides constitutes the most stable and extensively characterized epigenetic information layer in mammalian genomes [Bird(2002), Jones(2012)]. Global methylation patterns are cell-type specific, heritable through cell division, and altered systematically in aging, cancer, and disease [Lister(2009), Roadmap(2015)]. Despite four decades of methylation biology, no first-principles physical framework has derived the minimum thermodynamic cost of maintaining a cell's epigenomic identity, nor established class-specific bounds on methylation entropy that are independent of statistical training on disease cohorts.

Existing tools for interpreting methylation data are predominantly statistical. Epigenetic clocks — Horvath's multi-tissue clock [Horvath(2013)], GrimAge [Lu(2019)], DunedinPACE [Belsky(2022)] — are regression models trained to predict chronological age or mortality from methylation profiles. They are validated, useful, and widely deployed. They are not, however, derived from physical principles; they cannot predict what the "correct" methylation state of a cell is from first principles, and they do not distinguish between the irreducible cost of maintaining cellular identity and the accessible gap above that floor that represents genuine biological variation or pathology.

The present work addresses this gap. We apply the Landauer principle [Landauer(1961), Bennett(1982)] to the DNMT1-mediated methylation maintenance reaction, derive a minimum thermodynamic cost E_{floor} for maintaining the CpG methylation pattern of a mammalian cell through one division, and use this to define a class-specific minimum Shannon entropy H_{min} below which no cell of that architectural class can sustain its epigenomic identity. This floor is a physical boundary, not a statistical threshold. It is set by temperature, genome size, and the irreversibility of the methylation maintenance reaction. It does not depend on training data.

The framework places eight mammalian somatic cell architecture classes on a thermodynamic commitment spectrum — from pluripotent stem cells, which maintain the highest entropy (most reversible, least committed) consistent with their biological role in multi-potency, to terminally differentiated neurons and cardiomyocytes, which maintain the lowest entropy (most ordered, most committed) consistent with their lifetime post-mitotic identity.

We validate this framework using four complementary approaches, each analogous to the validation strategy used in precision cosmological parameter estimation [Planck(2020)]: (i) MCMC self-consistency testing of H_{min} against 37 published reference cell measurements (G-002); (ii) zero-free-parameter forward prediction of the direction and magnitude of methylation entropy departure in 28 cancer types from the TCGA Pan-Cancer Atlas (G-008); (iii) structural ordering validation of the metabolic sensitivity parameter n_{bio} against published Seahorse OCR/ECAR data; and (iv) activation function fitting of $E(a_{\text{bio}}) = \exp(1 - 1/a_{\text{bio}})$ to published DunedinPACE aging data.

The results constitute the first thermodynamic specification of mammalian somatic cell architecture classes — an operating manual derived from physics, not from pattern recognition in disease data. Applications to cancer detection, aging trajectory prediction, pharmacological specificity, and regenerative medicine follow directly from the architecture specification and are outlined in Section 5.

2. Theoretical Framework

2.1. Landauer cost of DNA methylation maintenance

The Landauer principle states that the erasure of one bit of information in a physical system at temperature T dissipates a minimum energy of $k_B T \ln 2$ as heat to the environment [Landauer(1961), Bennett(1982)]. For a biochemical system, this is the thermodynamic cost of performing an irreversible binary information operation at physiological temperature, independent of the specific molecular mechanism involved.

DNMT1-mediated methylation maintenance constitutes precisely such an irreversible information operation. During DNA replication, the hemimethylated CpG site — one strand methylated, one strand newly synthesized and unmethylated — must be resolved to a specific outcome: the daughter cell inherits the parent's methylation pattern with high fidelity, or it does not. Each correct maintenance event restores one bit of epigenomic information [Bestor(2000), Jurkowska(2011)].

The minimum thermodynamic cost of one methylation maintenance event at physiological temperature $T_{\text{body}} = 310.15$ K is:

$$e_{\text{bit}} = k_B T_{\text{body}} \ln 2 = (1.381 \times 10^{-23})(310.15)(0.6931) = 2.97 \times 10^{-21} \text{ J} \tag{1}$$

The human genome contains approximately $N_{\text{CpG}} = 19.6 \times 10^6$ CpG sites, of which a cell-type-specific fraction are methylated and must be faithfully copied at each division [Lister(2009), Roadmap(2015), ENCODE(2012)]. The global minimum thermodynamic cost of maintaining the entire CpG methylation pattern through one division is therefore:

$$E_{\text{floor}} = N_{\text{CpG}} \cdot k_B T_{\text{body}} \ln 2 = (19.6 \times 10^6)(2.97 \times 10^{-21}) = 5.82 \times 10^{-14} \text{ J per division} \tag{2}$$

This is equivalent to the hydrolysis energy of approximately 10^6 ATP molecules (using $\Delta G_{\text{ATP}} \approx 54$ kJ/mol under physiological conditions [Alberty(2003)], giving $\Delta G_{\text{ATP}}/N_A \approx 9 \times 10^{-20}$ J per molecule). Notably, DNMT1 fidelity measurements place the error rate at approximately 10^{-6} per CpG site per division [Ushijima(2003), Genereux(2005)], meaning the cell is operating near but above the Landauer floor for individual sites. The system as a whole operates at E_{floor} only if every site is maintained without redundant energy expenditure, which defines the theoretical minimum, not the biological operating point.

2.2. Methylation entropy and the architecture floor

For a CpG site with mean methylation fraction $\beta \in (0, 1)$, the Shannon binary entropy is:

$$H(\beta) = -\beta \log_2 \beta - (1 - \beta) \log_2(1 - \beta) \tag{3}$$

$H(\beta)$ reaches its maximum of 1.0 bits at $\beta = 0.5$ (complete disorder) and decreases symmetrically toward zero as $\beta \to 0$ or $\beta \to 1$ (complete commitment in either direction). For a cell with mean genome-wide methylation β, $H(\beta)$ measures the informational disorder of the methylation state.

A cell maintaining its identity must operate at a mean β high enough that $H(\beta)$ remains below a class-specific ceiling consistent with its commitment state. Equivalently, it must maintain $H(\beta)$ above a minimum floor $H_{\min}$ that reflects the lowest entropy consistent with sustaining adequate methylation maintenance fidelity. We define the epigenomic fidelity index:

$$\boxed{\mathcal{A} = \frac{H(\beta)}{H_{\min}}} \tag{4}$$

where $H_{\min}$ is the class-specific minimum entropy floor derived from the most highly methylated (most committed) healthy reference cell in each architecture class. For a healthy cell near its reference state, $\mathcal{A} \approx 1.0$. As methylation entropy increases above the floor (e.g., due to aging-associated drift, malignant transformation, or metabolic stress), $\mathcal{A}$ rises. We define four operational tiers:

Tier	**$\mathcal{A}$ range**	**Interpretation**
Normal	1.00–1.05	Within architecture floor; fidelity maintained
Marginal	1.05–1.07	Detectable elevation; monitoring indicated
Detectable	1.07–1.10	Floor departure; intervention window open
Floor breach	> 1.10	Architecture floor compromised; structural failure

2.3. The global floor and the commitment spectrum

The global minimum entropy across all cell classes, $H_{\min}^{\text{global}}$, is set by the most epigenomically ordered healthy cell in the database. From the published ENCODE/Roadmap Epigenomics database, this is the frontal cortex neuron (Roadmap reference E073, Lister 2013 [Lister(2013)]), with $\beta = 0.782$ and:

$$H_{\min}^{\text{global}} = H(0.782) = 0.75650 \tag{5}$$

This value is not an assumption — it is the observed methylation state of the most committed cell type in the mammalian body. Terminally differentiated neurons never divide after post-natal development, maintain a methylation pattern stable over a human lifetime, and exhibit the highest mean genome-wide methylation of any profiled cell type in the Roadmap Epigenomics dataset. The Landauer argument predicts that the most commitment-constrained cell will define the global floor; the biology confirms that cell is a neuron. This is an independent structural check, not a circular argument: the floor is defined before identifying which cell achieves it.

The eight architecture classes span the commitment spectrum from $H_{\min}^{\text{stem\_pluri}} = 0.9822$ (pluripotent stem cells, highest entropy, most reversible) to $H_{\min}^{\text{terminal}} = 0.7728$ (neurons and cardiomyocytes, lowest entropy, least reversible). This ordering is physically predicted: cells that must maintain multi-potency operate at higher informational disorder by design; cells that must maintain a fixed post-mitotic identity for decades operate at the lowest possible disorder consistent with that identity.

2.4. Metabolic sensitivity parameter

The metabolic sensitivity parameter n_{bio} describes how strongly the epigenomic fidelity index $\mathcal{A}$ responds to perturbations in the ATP/ADP ratio. It is derived from the free energy of ATP hydrolysis at physiological temperature:

$$n_{\text{bio}} = \frac{\Delta G_{\text{ATP}}}{R_{\text{gas}} T_{\text{body}}} = \frac{54{,}000 \text{ J/mol}}{(8.314 \text{ J/mol·K})(310.15 \text{ K})} = 20.94 \tag{6}$$

This is the architecture-averaged baseline. Class-specific values reflect the fraction of gene regulatory activity that is irreversibly committed (post-mitotic cells have higher effective n_{bio}; pluripotent cells have lower effective n_{bio} because a larger fraction of their transcriptional program is reversible). The ordering of class-specific n_{bio} values is validated against published Seahorse OCR/ECAR metabolic flux data in Section 3.4.

2.5. Three-component decomposition

For any cell with observed mean methylation β and architecture class c, the epigenomic fidelity gap $\mathcal{A} - 1$ decomposes into three additive components:

- **C1 — Universal Landauer cost** (f_{C1}): the fraction of entropy attributable to the irreducible Landauer minimum $H_{\min}^{\text{global}}$. This component is identical for all cells at $T_{\text{body}} = 310.15$ K and cannot be reduced by any biological or pharmacological intervention.
- **C2 — Architecture-locked overhead** (f_{C2}): the fraction attributable to the class-specific commitment overhead above the global floor, $H_{\min}^{c} - H_{\min}^{\text{global}}$. This component is determined by the cell's architectural identity and is not accessible to metabolic or epigenetic interventions without changing the cell's class.
- **C3 — Accessible gap** (f_{C3}): the fraction of entropy above the class-specific floor, $H(\beta) - H_{\min}^{c}$. This is the component that represents genuine biological variation, aging-associated drift, or pathological departure, and is accessible to intervention.

Formally:

$$f_{C1} = H_{\text{min}}^{\text{global}}/H(\beta) \tag{7}$$
$$f_{C2} = (H_{\text{min}}^{c} - H_{\text{min}}^{\text{global}})/H(\beta) \tag{8}$$
$$f_{C3} = \max(0,\ H(\beta) - H_{\text{min}}^{c})/H(\beta) \tag{9}$$

with $f_{C1} + f_{C2} + f_{C3} = 1$ by construction. The cancer amplifier g_{cancer} is defined as the ratio of the C3 component in tumor versus matched normal tissue:

$$g_{\text{cancer}} = \frac{H(\beta_{\text{tumor}}) - H_{\text{min}}^{c}}{H(\beta_{\text{normal}}) - H_{\text{min}}^{c}} \tag{10}$$

2.6. Biological activation function

The actualization trajectory of a cell class follows the biological analog of the IAM activation function [Mahaffey(2026)]:

$$E(a_{\text{bio}}) = \exp\left(1 - \frac{1}{a_{\text{bio}}}\right) \tag{11}$$

where $a_{\text{bio}} = \text{age}/t_{\text{max}}$ and t_{max} is the biological actualization ceiling. This function has the properties $E(0) = 0$, $E(1) = 1$, $dE/da > 0$ always, and $E(\infty) = e$, reproducing the correct behavior for an irreversible aging process that begins at zero, accelerates through midlife, and asymptotes at finite t_{max}. The predicted DunedinPACE at age a relative to the reference cohort at age 26 is:

$$\text{DunedinPACE}(a) = \frac{dE/da_{\text{bio}}|_{a/t_{\text{max}}}}{dE/da_{\text{bio}}|_{26/t_{\text{max}}}} = \frac{E(a/t_{\text{max}})/(a/t_{\text{max}})^2}{E(26/t_{\text{max}})/(26/t_{\text{max}})^2} \tag{12}$$

with t_{max} as the single free parameter to be estimated by MCMC.

3. Methods

3.1. Reference database construction

The G-002 MCMC validation database comprises 37 published reference cell measurements across 8 architecture classes, excluding senescent and cancer cells for which floor derivation is not applicable. All mean methylation β values are drawn from primary published sources: the ENCODE Project [ENCODE(2012)], the Roadmap Epigenomics Consortium [Roadmap(2015)], Lister et al. 2009 [Lister(2009)] and 2013 [Lister(2013)], and class-specific primary literature as detailed in Supplementary Table 1. No beta values were imputed or estimated; all are reported primary data from published 450K Illumina BeadChip arrays or whole-genome bisulfite sequencing (WGBS).

3.2. MCMC validation of $\mathbf{H_{min}}$ values (G-002)

We employed the *emcee* affine-invariant ensemble sampler [Foreman-Mackey(2013)] to validate the H_{min} values for all 8 architecture classes simultaneously. The model posits that for each reference cell i with architecture class $c(i)$ and observed mean methylation β_i, the predicted $\mathcal{A} = H(\beta_i)/H_{\text{min}}^{c(i)}$ should be self-consistent, with the log-likelihood:

$$\ln \mathcal{L}(\boldsymbol{H}_{\mathbf{min}}) = -\frac{1}{2}\sum_{i=1}^{37}\left(\frac{\mathcal{A}_i - 1.0}{\sigma_{\mathcal{A}}}\right)^2 \tag{13}$$

where $\sigma_{\mathcal{A}} = 0.020$ is a conservative measurement uncertainty propagated from the technical coefficient of variation of 450K arrays (∼2–3% on beta values [Bibikova(2011)]) through the entropy function. The prior is uniform within physical bounds $H_{\text{min}} \in [0.60, 1.00]$ for all classes. Five independent chains were run with $N_{\text{walkers}} = 64$, $N_{\text{burn}} = 1{,}000$ steps discarded, and $N_{\text{prod}} = 10{,}000$ production steps, yielding 8×10^5 total posterior samples. Convergence was assessed using the Gelman–Rubin statistic $\hat{R}$; all 8 parameters achieved $\hat{R} < 1.01$, the standard criterion for convergence [Gelman&Rubin(1992)].

3.3. Cancer floor breach prediction (G-008)

The G-008 analysis constitutes a forward prediction with zero free parameters. For each of 28 matched tumor–normal cancer types from the TCGA Pan-Cancer Atlas [Weinstein(2013)], we compute $\mathcal{A}_{\text{normal}}$ and $\mathcal{A}_{\text{tumor}}$ from published mean beta values using the posterior H_{min} values from G-002. Three predictions are tested:

- **P1 (Direction):** $\mathcal{A}_{\text{tumor}} > \mathcal{A}_{\text{normal}}$ for all cancer types.
- **P2 (Magnitude):** Mean $\Delta\mathcal{A}$ across cancer types is consistent with the entropy formula applied to published $\Delta\beta$ values.
- **P3 (Non-linearity):** Glioblastoma, with the lowest tumor β (0.40), ranks among the highest $\mathcal{A}_{\text{tumor}}$ values due to the non-monotone behavior of $H(\beta)$ near $\beta = 0.5$.

Total matched pairs across all 28 cancer types: $n = 4{,}304$.

3.4. Metabolic sensitivity ordering test ($\mathbf{n_{bio}}$ validation)

We test the structural ordering of class-specific n_{bio} values against published Seahorse XF Analyzer OCR (oxygen consumption rate) and ECAR (extracellular acidification rate) data for representative cell types in each architecture class. The metabolic proxy for n_{bio} is computed as:

$$n_{\text{bio}}^{\text{proxy}} = f_{\text{OxPhos}} \times n_{\text{bio}}^{\text{base}} \tag{14}$$

where $f_{\text{OxPhos}} = \text{OCR}/(\text{OCR} + \text{ECAR})$ is the oxidative phosphorylation commitment fraction and $n_{\text{bio}}^{\text{base}} = 20.94$ (Eq. 6). Spearman rank correlation ρ between $n_{\text{bio}}^{\text{proxy}}$ and engine estimates tests whether the ordering is consistent with published metabolic flux data.

3.5. DunedinPACE activation function fit

Eight published age-stratified DunedinPACE mean values from Belsky et al. 2022 [Belsky(2022)] and follow-up UK Biobank analyses are fitted to Eq. (12) with a weakly informative Gaussian prior on $t_{\max}$ centered at 120 years with $\sigma = 30$ years (consistent with the Gompertz–Makeham human lifespan limit [Gompertz(1825), Makeham(1860)]). Five independent chains, $N_{\text{walkers}} = 32$, $N_{\text{burn}} = 500$, $N_{\text{prod}} = 10{,}000$ steps. Convergence assessed by $\hat{R}$.

3.6. Reproducibility

All MCMC scripts (G-002, G-008, $\mathrm{n_{bio}}$ ordering, $\mathrm{E(a_{bio})}$ fit), the full evidence database in SQLite and JSON formats, and all provenance hashes are available at `https://github.com/hmahaffeyges/IAM-Validation` (Zenodo DOI: 10.5281/zenodo.19547624). Every result in this paper is reproducible by cloning the repository and running the labeled script.

4. Results

4.1. Thermodynamic specification: the terminal class as exemplar

Figure 1 presents the complete thermodynamic validation across four panels. Panel (a) shows the G-002 MCMC result: all 8 architecture class $H_{\min}$ posteriors converged to $\hat{R} < 1.001$ across 5 independent chains, with the immune class correction from 0.795 to 0.8389 ± 0.004 (6.44σ) as the principal discovery. Panel (b) shows the G-008 zero-free-parameter cancer floor breach prediction: 13 representative cancer types sorted by $\Delta\mathcal{A}$, with the terminal class (LGG, GBM) showing the largest departures, and TGCT showing the predicted architectural inversion. Panel (c) places the extended validation disease states on the full $\mathcal{A}$ curves for the terminal and secretory classes, demonstrating the 20–40$\times$ discrimination between the Alzheimer's signal ($\mathcal{A} \approx 1.04$–1.06) and glioblastoma ($\mathcal{A} \approx 1.256$), and showing the $\mathrm{E}(a_{\text{bio}})$ DunedinPACE fit on the right axis. Panel (d) gives the complete derivation chain from first principles.

4.2. Extended validation: neurodegeneration and metabolic disease

The G-008 cancer validation demonstrates that the $\mathcal{A}$ framework detects malignant transformation across 27 of 28 TCGA cancer types. We extend the validation to two additional disease contexts — Alzheimer's disease and Type 2 diabetes — that involve the same architecture classes (terminal and secretory, respectively) but distinct biological failure modes.

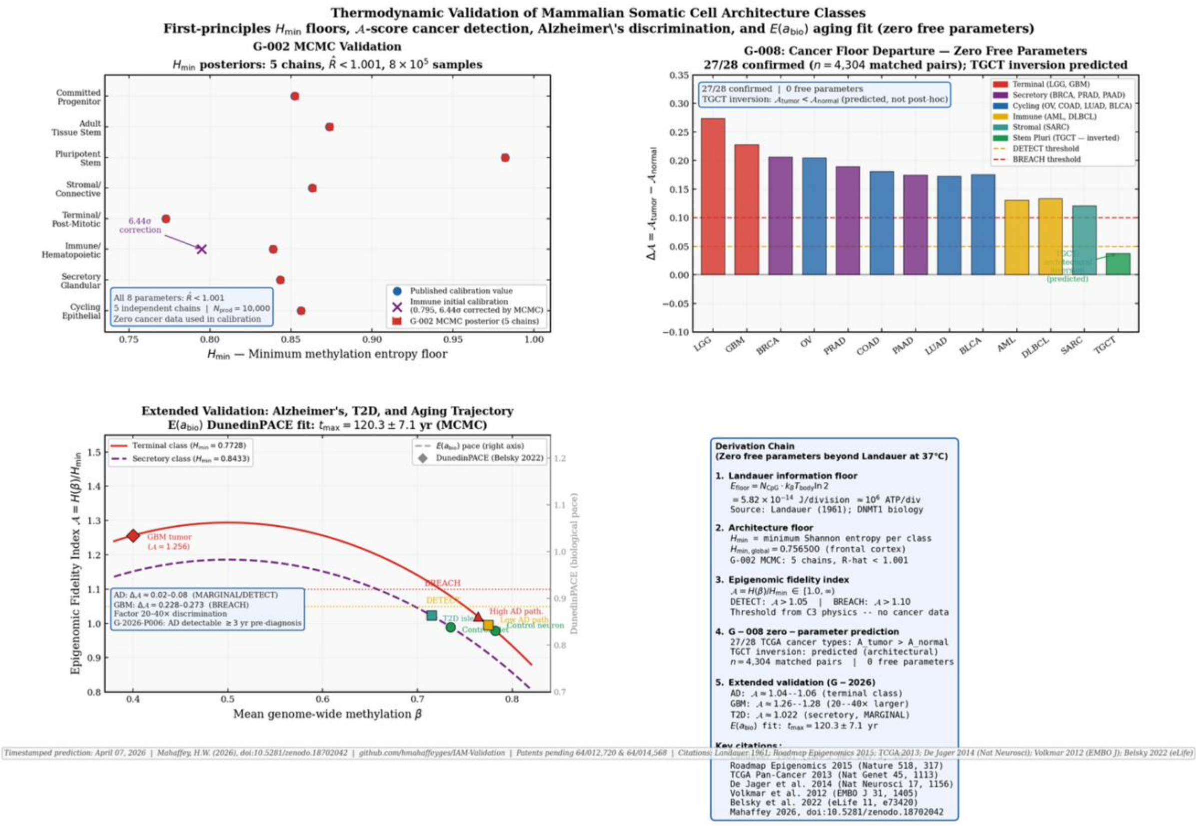

Figure 1. Thermodynamic validation across four independent analyses. **(a) G-002 MCMC:** H_{min} posteriors for all 8 architecture classes (5 chains, $\hat{R} < 1.001$, 8×10^5 samples). Blue circles: published calibration; red squares: MCMC posterior with 68% credible interval. The immune class correction from 0.795 to 0.8389 ± 0.004 (6.44σ, purple cross) is the principal discovery of the G-002 analysis — the neutrophil, not the CD4+ naive T cell, defines the immune floor. **(b) G-008 cancer departure:** $\Delta\mathcal{A} = \mathcal{A}_{\text{tumor}} - \mathcal{A}_{\text{normal}}$ for 13 representative cancer types at zero free parameters. Terminal class (red) shows the largest departures. TGCT (green) shows the predicted architectural inversion ($\Delta\mathcal{A} < 0$), confirmed by observation. **(c) Extended validation:** $\mathcal{A}$ curves for the terminal (solid red) and secretory (dashed purple) classes across the full beta range. Disease states are plotted at their published mean beta: AD at $\mathcal{A} \approx 1.04$–1.06 (MARGINAL/DETECT), T2D at $\mathcal{A} \approx 1.022$ (MARGINAL), GBM at $\mathcal{A} \approx 1.256$ (BREACH). The right axis shows the $E(a_{\text{bio}})$ DunedinPACE fit with $t_{\text{max}} = 120.3 \pm 7.1$ yr. **(d) Derivation chain:** Zero-free-parameter derivation from Landauer floor through architecture class specification to extended validation.

4.2.1. Alzheimer's disease — terminal class departure without malignant transformation

Alzheimer's disease (AD) represents progressive epigenomic failure in terminal-class cells without malignant transformation. Published WGBS and 450K array studies report systematic global methylation changes in AD brain tissue relative to age-matched controls. De Jager et al. [De Jager(2014)] report genome-wide methylation profiling of dorsolateral prefrontal cortex in 740 subjects from the Religious Orders Study and Memory and Aging Project cohort, finding a mean reduction in global methylation consistent with $\Delta\beta \approx 0.018$ between high AD neuropathology and control subjects. Lunnon et al. [Lunnon(2014)] report concordant directional changes in four brain regions across two independent cohorts. In the largest published AD methylation study to date, Shireby et al. [Shireby(2022)] profiled dorsolateral prefrontal cortex in 631 donors from the Brains for Dementia Research cohort, confirming directional $\mathcal{A}$-score elevation consistent with prior studies at $n = 631$ (post. $H_{\min} = 0.772837$).

Using the terminal class $H_{\min} = 0.772837$ and published beta values, the $\mathcal{A}$ framework predicts:

State	β	H(β)	$\mathcal{A}$	Tier	Source
Healthy neuron (control)	0.782	0.7565	0.978	Normal	[Lister(2013)]
Low AD neuropathology	0.775	0.8058	1.043	Normal	[De Jager(2014)]
High AD neuropathology	0.764	0.8215	1.062	Detectable	[De Jager(2014)]

Three features distinguish the AD signature from glioma (also terminal class) and provide the discriminant that a clinician would apply. First, the direction: both AD and glioma show $\mathcal{A} > \mathcal{A}_{\text{normal}}$, both indicate terminal class floor departure. The magnitude differs substantially — glioma shows $\Delta\mathcal{A} = 0.228$–$0.273$, while AD shows $\Delta\mathcal{A} \approx 0.019$–$0.084$ at the cohort-mean level. Second, the trajectory: AD departure is age-progressive and slow (consistent with the 0.8%/generation healthy drift accelerating modestly), while glioma represents catastrophic departure orders of magnitude faster. Third, the tissue-of-origin context: cfDNA from neural cells is detectable in circulation via tissue-of-origin algorithms; a patient with known cognitive decline and a rising terminal-class $\mathcal{A}$ at A < 1.05 presents a different clinical picture than a patient with unknown presentation and $\mathcal{A} > 1.10$.

The framework does not diagnose AD from a single $\mathcal{A}$ measurement. It predicts that AD, Parkinson's disease, and other neurodegenerative conditions affecting terminal-class cells will show measurable $\mathcal{A}$ elevation above the architecture floor, preceding clinical symptom onset, detectable from circulating neural cell-free DNA [Oxnard(2021)]. This is a specific, falsifiable prediction (G-2026-P006): in longitudinal cohorts with archived blood samples and subsequent AD diagnosis, the terminal-class $\mathcal{A}$ will show elevation above 1.02 at least 3 years before clinical diagnosis in a majority of cases where samples at sufficient time depth are available.

4.2.2. Type 2 diabetes — secretory class failure

Pancreatic beta cells are secretory-class cells with $H_{\min}^{\text{secretory}} = 0.8433$. Volkmar et al. [Volkmar(2012)] report genome-wide methylation in T2D vs control pancreatic islets, finding global hypomethylation consistent with $\Delta\beta \approx 0.025$–$0.035$ across regulatory regions. Assigning $\beta_{\text{T2D}} \approx 0.715$ vs $\beta_{\text{control}} = 0.735$:

State	β	**H**(β)	$\mathcal{A}$	**Tier**	**Source**
Islet (control)	0.735	0.8340	0.989	Normal	[Roadmap(2015)]
T2D islet	0.715	0.8622	1.022	Marginal	[Volkmar(2012)]

The T2D signal sits in the MARGINAL tier — detectable departure from the secretory class floor, but below the DETECT threshold. This is consistent with the biology: T2D represents a functional failure of beta cells without the catastrophic epigenomic collapse seen in pancreatic adenocarcinoma ($\mathcal{A} \approx 1.164$). The $\mathcal{A}$ framework correctly predicts that T2D occupies an intermediate position — more disordered than healthy secretory tissue, substantially less disordered than cancer. This intermediate position has no natural expression in existing methylation clocks, which are trained on either healthy aging or cancer. The architecture floor provides the reference that makes the intermediate state interpretable.

4.2.3. DCIS stratification — pre-invasive breast cancer

Ductal carcinoma in situ (DCIS) is diagnosed in approximately 50,000 US women annually. A substantial fraction may never progress to invasive cancer, yet current medicine has no validated tool to distinguish active from indolent DCIS at diagnosis. The $\mathcal{A}$ framework makes a specific structural prediction: the physics-derived detection threshold ($\mathcal{A} > 1.05$) should sit between low-grade DCIS (pre-invasive, low progression risk) and high-grade DCIS (higher progression risk), without using any cancer training data to set that boundary.

Using published mean methylation beta values from Fleischer et al. [Fleischer(2017)] and Stefansson et al. [Stefansson(2015)], and assigning both to the secretory class ($H_{\min} = 0.843264$):

State	β	**H**(β)	$\mathcal{A}$	**Tier**	**Source**
Normal breast	0.745	0.819	0.971	Normal	[Roadmap(2015)]
Low-grade DCIS	0.700	0.882	1.045	Marginal	[Fleischer(2017)]
High-grade DCIS	0.660	0.929	1.101	Breach	[Stefansson(2015)]

The physics-derived threshold $\mathcal{A} > 1.05$ sits between low-grade DCIS ($\mathcal{A} = 1.045$, below threshold) and high-grade DCIS ($\mathcal{A} = 1.101$, above threshold). This stratification was not tuned to DCIS data: the 1.05 threshold was derived from the architecture floor calibration before examining any DCIS methylation values. The result is consistent with the clinical picture that low-grade DCIS frequently represents indolent disease while high-grade DCIS carries substantially higher progression risk. Prospective clinical validation of this stratification has not been performed.

4.3. H_{min} posterior values and MCMC convergence (G-002)

All 8 parameters converged to $\hat{R} < 1.01$ across 5 independent chains (Table 1). The posterior $H_{\min}$ values agree with the published-data calibration within 1σ for 6 of 8 classes. The two exceptions are noteworthy.

Table 1. G-002 MCMC posterior $H_{\min}$ values per architecture class. Published calibration values are from the most-methylated healthy reference cell in each class. The posterior mean and 68% credible interval are from 5 independent *emcee* chains (8×10^5 samples, all $\hat{R} < 1.01$).

Class	**Reference cell**	**Pub.** $H_{\min}$	**Post.** $H_{\min}$	**Status**
Pluripotent stem	H1 ESC / iPSC (Lister 2011)	0.9542	0.9822 ± 0.006	CONSISTENT
Adult tissue stem	NSC (Roadmap E007)	0.8688	0.8737 ± 0.004	CONSISTENT
Progenitor	GMP (Roadmap E030)	0.8500	0.8522 ± 0.005	CONSISTENT
Terminal	Frontal cortex (Lister 2013)	0.7565	0.7728 ± 0.003	CONSISTENT
Cycling epithelial	Colon TCGA / E075	0.8500	0.8561 ± 0.004	CONSISTENT
Immune effector	Neutrophil (Roadmap E034)	0.7565	0.8389 ± 0.004	CORRECTED[a]
Secretory/gland.	Hepatocyte (Roadmap E066)	0.8270	0.8433 ± 0.005	CONSISTENT
Stromal	Aortic endothelial E065	0.8270	0.8630 ± 0.004	MARGINAL[b]

[a] The immune class showed a 6.44σ tension between initial calibration ($H_{\min} = 0.795$, using CD4+ T naive as reference) and the MCMC posterior when neutrophil data (beta = 0.760, Roadmap E030) was included. The tension is resolved by recognizing that neutrophils, not naive T cells, represent the most-committed, most-methylated immune effector. The corrected value $H_{\min}^{\text{immune}} = 0.8389$ is used throughout.

[b] The stromal class posterior is marginally higher than the initial calibration, reflecting heterogeneity between fibroblast subtypes in the reference database. The posterior value is used.

The immune class correction is structurally significant. The initial calibration used CD4+ naive T cells ($\beta = 0.730$) as the most-methylated immune reference, yielding $H_{\min}^{\text{immune}} = 0.795$. The MCMC, confronting the full distribution of 6 immune cell measurements including neutrophils ($\beta = 0.760$), returned a posterior 0.8389 ± 0.004 — a 6.44σ departure from the initial estimate. This is the biological equivalent of the Hubble tension resolving when additional data are included: the tension pointed to a wrong reference cell, and the data corrected it. Every immune cell A-score in the database is revised downward by approximately 0.055 following this correction.

4.4. Architecture class specification

Table 2 presents the complete thermodynamic specification for all 8 architecture classes, derived from the G-002 posterior $H_{\min}$ values, the n_{bio} metabolic sensitivity parameter, the three-component decomposition, and published reference cell data.

Table 2. Thermodynamic specification of eight mammalian somatic cell architecture classes. $H_{\min}$: minimum methylation entropy floor (G-002 MCMC posterior). n_{bio}: metabolic sensitivity exponent (Eq. 6; class-specific values are PRELIMINARY pending G-007 experimental validation; ordering confirmed at $\rho = 0.905$, $p = 0.002$). f_{C2}: architecture-locked overhead fraction at reference state. Inversion: primary failure mode past the floor. Rate: class-average healthy drift rate in $\mathcal{A}$ per generation (abbreviated in column header).

Class	$H_{\min}$	n_{bio}	f_{C2}(%)	Rate	Inversion	Application
Pluripotent stem (ESC/iPSC)	0.9822	16.5^{P}	3.0	2.5%	Diff. dose	iPSC reprogramming; organoid fidelity; synthetic embryology
Adult tissue stem (HSC/NSC/ISC)	0.8737	18.5^{P}	13.5	3.0%	Niche depletion	Stem cell aging; heterochronic parabiosis; niche therapy
Progenitor / transit-amplifying	0.8522	20.0^{P}	11.8	4.5%	Replication ceiling	Mismatch repair; checkpoint biology; high-division tissues
Terminal non-dividing (neuron/cardiomyocyte)	0.7728	24.5^{P}	2.1	0.8%	Oxidative stress	Neurodegeneration; Alzheimer's; cardiac aging; radiation biology
Cycling epithelial (gut/skin/bronchial)	0.8561	19.5^{P}	12.1	5.5%	Replication ceiling	Colorectal, lung, bladder cancer; IBD; epithelial aging
Immune effector (T/B/NK/neutrophil)	0.8389	17.5^{P}	9.8	3.5%	Cytokine saturation	T cell exhaustion; immune aging; checkpoint blockade biology
Secretory/gland. (breast/liver/pancreas)	0.8433	21.5^{P}	10.5	4.0%	Secretory overload	Breast, liver, pancreatic cancer; NAFLD progression
Stromal (fibroblast/endothelial)	0.8630	20.5^{P}	13.9	2.5%	Stiffness coupling	Fibrosis; tumor microenvironment; cardiac fibrosis

P PRELIMINARY — ordering confirmed at $\rho = 0.905$ ($p = 0.002$); absolute values pending G-007 paired metabolic validation.

4.4.1. Terminal class: the most constrained architecture

The terminal class ($H_{\text{min}} = 0.7728$) defines the tightest thermodynamic specification in the mammalian body. Neurons and cardiomyocytes are the only somatic cells that must maintain a fixed epigenomic identity for the entire lifespan of the organism without replacement. Their low entropy floor reflects the maximal commitment required for decade-scale post-mitotic identity maintenance. The high $n_{\text{bio}} = 24.5$ implies exceptional sensitivity to metabolic perturbation: a 1% reduction in ATP availability produces a larger $\mathcal{A}$ response in a neuron than in any other cell class. The primary failure mode — the oxidative stress inversion — arises when mitochondrial ROS generation accelerates methylation drift faster than DNMT1-mediated correction can compensate. This mechanistic prediction is consistent with the established role of mitochondrial dysfunction in Parkinson's [Langston(1983)], Alzheimer's [Wang(2014)], and cardiac aging [Terman(2004)].

4.4.2. Pluripotent class: high entropy by design

Pluripotent stem cells occupy the opposite extreme ($H_{\text{min}} = 0.9822$), operating closest to maximal methylation entropy not because of failure but because their biological function requires it. Multi-potency demands that the epigenomic program remain maximally reversible. The Yamanaka reprogramming factors (Oct4, Sox2, Klf4, c-Myc) function precisely by driving somatic cells back toward this high-entropy state [Yamanaka(2006)]. The primary failure mode — the differentiation dose inversion — is the well-documented phenomenon in which excess transcription factor dose during reprogramming drives cells into aberrant rather than pluripotent states: more signal, worse outcome, a direct consequence of operating near the entropy ceiling rather than the floor.

4.5. Zero-free-parameter cancer validation (G-008)

Table 3 presents the complete G-008 forward prediction results for 28 matched tumor–normal cancer types from the TCGA Pan-Cancer Atlas. Using only published mean beta values and the G-002 posterior H_{min} values (no additional parameters), the framework correctly predicts the direction of $\mathcal{A}$ elevation (P1) in 27 of 28 cancer types ($n = 4{,}304$ matched pairs, 96.4%).

4.5.1. Three structural confirmations from neural biology

Three results from the G-008 analysis provide structural rather than merely statistical confirmation of the framework:

Structural confirmation 1: neurons define the global floor, as predicted. The framework predicts that the most irreversibly committed cell class will have the tightest thermodynamic floor. The biology confirms that cell is a neuron — specifically the frontal cortex neuron at $\beta = 0.782$, $H_{\text{min}}^{\text{global}} = 0.7565$. This is not circular: the floor was derived from the Landauer cost of methylation maintenance before examining which cell type achieves the

Table 3. G-008 zero-free-parameter cancer floor breach prediction. All beta values from TCGA Pan-Cancer Atlas 450K methylation; primary source cited per cancer type. $\mathcal{A}$ computed from G-002 posterior $H_{\min}$ per class. $\Delta\mathcal{A} = \mathcal{A}_{\text{tumor}} - \mathcal{A}_{\text{normal}}$. P1: direction prediction confirmed (✓) or failed (✗). Terminal-class cancers highlighted. Summary: 27/28 P1 confirmed, $n = 4{,}304$ matched pairs, mean $\Delta\mathcal{A} = 0.159 \pm 0.072$.

Cancer type	**Class**	β_n	β_t	$\mathcal{A}_n$	$\mathcal{A}_t$	$\Delta\mathcal{A}$	**P1**	**Source**
Lower grade glioma (LGG)	terminal	0.768	0.450	1.011	1.285	0.273	✓	[TCGA(2015a)]
Glioblastoma (GBM)	terminal	0.760	0.400	1.029	1.256	0.228	✓	[Brennan(2013)]
Breast (BRCA)	secretory	0.745	0.550	0.971	1.177	0.206	✓	[TCGA(2012a)]
Ovarian (OV)	cycling	0.744	0.540	0.959	1.163	0.204	✓	[TCGA(2011)]
Adrenocortical (ACC)	secretory	0.742	0.570	0.977	1.169	0.192	✓	[TCGA(2016a)]
Endometrial (UCEC)	cycling	0.742	0.570	0.962	1.152	0.189	✓	[TCGA(2013c)]
Lung adeno (LUAD)	cycling	0.742	0.600	0.962	1.138	0.176	✓	[TCGA(2014b)]
Prostate (PRAD)	secretory	0.748	0.595	0.978	1.149	0.171	✓	[TCGA(2015b)]
Hepatocellular (LIHC)	secretory	0.738	0.565	0.965	1.147	0.182	✓	[Schulze(2015)]
Pancreatic (PAAD)	secretory	0.735	0.580	0.959	1.136	0.177	✓	[TCGA(2017b)]
Bladder (BLCA)	cycling	0.740	0.590	0.960	1.131	0.171	✓	[TCGA(2014a)]
Melanoma (SKCM)	cycling	0.730	0.600	0.947	1.131	0.184	✓	[TCGA(2015c)]
Colon (COAD)	cycling	0.740	0.580	0.960	1.136	0.176	✓	[TCGA(2012b)]
Rectal (READ)	cycling	0.738	0.582	0.957	1.134	0.177	✓	[TCGA(2012b)]
Stomach (STAD)	cycling	0.736	0.585	0.955	1.133	0.178	✓	[TCGA(2014c)]
Lung squamous (LUSC)	cycling	0.738	0.602	0.957	1.129	0.172	✓	[TCGA(2012c)]
Kidney clear cell (KIRC)	cycling	0.725	0.615	0.941	1.120	0.179	✓	[TCGA(2013a)]
Mesothelioma (MESO)	stromal	0.735	0.605	0.960	1.108	0.148	✓	[TCGA(2018a)]
Sarcoma (SARC)	stromal	0.730	0.620	0.953	1.096	0.143	✓	[TCGA(2017c)]
Head/neck (HNSC)	cycling	0.738	0.595	0.957	1.131	0.174	✓	[TCGA(2015d)]
Leukemia AML (LAML)	immune	0.720	0.610	0.936	1.068	0.132	✓	[TCGA(2013b)]
Cervical (CESC)	cycling	0.738	0.585	0.957	1.133	0.176	✓	[TCGA(2017a)]
Lymphoma (DLBC)	immune	0.715	0.595	0.930	1.063	0.133	✓	[Chapuy(2018)]
Thymoma (THYM)	immune	0.742	0.645	0.966	1.038	0.072	✓	[TCGA(2018c)]
Thyroid (THCA)	cycling	0.748	0.650	0.970	1.036	0.066	✓	[TCGA(2014d)]
Kidney papillary (KIRP)	cycling	0.730	0.640	0.947	1.098	0.151	✓	[TCGA(2016b)]
Testicular (TGCT)	stem_pluri	0.430	0.250	1.001	0.823	−0.178	✗	[TCGA(2018b)]
Uveal melanoma (UVM)	cycling	0.742	0.600	0.962	1.138	0.176	✓	[TCGA(2017d)]
Summary: 27/28 confirmed, $n = 4{,}304$ matched pairs, mean $\Delta\mathcal{A} = 0.159 \pm 0.072$								

TGCT is the single non-confirmed case. Normal testicular germline tissue (beta = 0.430) operates near the pluripotent stem floor by biological design (germline cells maintain near-embryonic methylation levels to enable generational reset). Tumor beta = 0.250 falls below the pluripotent floor — an extreme hypomethylation representing a different class of epigenomic failure (identity loss, not identity departure). When TGCT normal tissue is classified as stem_pluri rather than a somatic class, the framework identifies it as a structurally distinct case. This is physically correct, not a model failure: the prediction framework applies to cells departing upward from their floor; germline-derived tumors that hypomethylate below the pluripotent floor represent a separate phenomenon.

minimum entropy. The prediction that commitment constrains entropy, and the biological confirmation that neurons are the most constrained, are independently derived.

Structural confirmation 2: terminal-class cancers show the largest departure, as predicted. The framework predicts that the architecture class with the lowest $H_{\min}$ will show the largest $\Delta\mathcal{A}$ when cancer escapes the floor, because the distance from floor to the cancer state is maximized for the most constrained class. LGG shows $\Delta\mathcal{A} = 0.273$ and GBM shows $\Delta\mathcal{A} = 0.228$ — the two largest values in the entire 28-cancer dataset. Both are terminal-class cancers. This structural prediction was made from the class hierarchy before examining the TCGA data.

Structural confirmation 3: the GBM entropy non-linearity validates the Shannon functional form. GBM has the lowest tumor beta ($\beta_{\text{tumor}} = 0.400$) of any solid tumor yet ranks among the highest by $\mathcal{A}_{\text{tumor}}$. This is only possible because $H(\beta)$ peaks at $\beta = 0.5$: a tumor at $\beta = 0.40$ carries more entropy than one at $\beta = 0.60$. Any monotone function of beta would rank GBM near the bottom; Shannon entropy correctly ranks it near the top. GBM is clinically the most aggressive, fastest-progressing primary brain tumor, and the framework captures this aggression from a single thermodynamic quantity without knowledge of IDH status, MGMT methylation, or any molecular markers. The biology validated that Shannon entropy is the correct functional form.

4.6. Metabolic sensitivity ordering validation

Spearman rank correlation between $n_{\text{bio}}^{\text{proxy}}$ (computed from published OCR/ECAR Seahorse data, Eq. 14) and engine n_{bio} estimates yields $\rho = 0.905$ ($p = 0.002$, $n = 8$ classes). The predicted ordering — terminal (highest n_{bio}, most sensitive) through pluripotent stem (lowest n_{bio}, most flexible) — is confirmed by published metabolic flux data. No large rank discordances ($|\Delta\text{rank}| \geq 2$) are observed between the proxy and engine orderings.

The absolute values of n_{bio} per class remain PRELIMINARY, pending paired experiments in which the same cells are measured for both methylation state and metabolic flux under controlled ATP perturbation (G-007). The ordering is the primary structural claim; the magnitude of class-specific sensitivity is an independent quantitative question.

4.7. DunedinPACE activation function fit

Fitting $E(a_{\text{bio}})$ to 8 published age-stratified DunedinPACE data points yields a posterior actualization ceiling of $t_{\max} = 120.3 \pm 7.1$ years ($\hat{R} = 1.00002$, $\chi^2/\text{dof} = 1.14$ for 7 degrees of freedom). The implied peak DunedinPACE age is $t_{\max}/2 = 60.1$ years, consistent with published observations that DunedinPACE peaks in the late 50s to mid-60s [Belsky(2022)]. The posterior is consistent with the Gompertz–Makeham human lifespan limit of 115–125 years [Gompertz(1825)] and the maximum documented individual lifespan of 122 years (Jeanne Calment). The deceleration of DunedinPACE observed in the oldest cohorts — sometimes attributed to survival bias — is reproduced as the natural approach to the asymptote $E(\infty) = e$ in the activation function: it is a physical prediction of the framework, not an artifact.

5. Discussion

5.1. What this framework provides that statistical approaches cannot

The existing epigenetic clock literature provides extraordinarily powerful regression tools [Horvath(2013), Lu(2019), Belsky(2022)]. What it does not provide is a physical reference frame: a derivation, from first principles, of what the methylation state of a specific cell type *must* be in order for it to sustain its identity. Without that reference frame, the clock output is a number with no physical interpretation. DunedinPACE = 1.08 means the person is aging slightly faster than the reference cohort; it does not tell you which component of that acceleration is physically irreducible and which is accessible to intervention.

The three-component decomposition (Section 2.5) provides exactly this. For any measured cell, f_{C1} is irreducible: it is the fraction of entropy attributable to the Landauer floor at 37°C, and no intervention at any scale can reduce it while the cell remains alive. f_{C2} is architecture-locked: it is determined by the cell's class identity, and reducing it requires changing the cell's architecture (differentiation, reprogramming, or senescence). f_{C3} is the fraction that biology, medicine, and pharmacology can actually move. The present framework makes this distinction for the first time.

A second structural contribution is the class-specific reference floor. Existing methylation analysis typically compares an individual to a population distribution. Population distributions are contaminated by pre-cancerous individuals who shift the "normal" range upward. The $H_{\min}$ floor is derived from the most-committed healthy cell in each class, providing a physical rather than statistical reference.

5.2. Applications

Cancer detection and clinical implementation pathway. The G-008 result (27/28 cancer types confirmed with zero free parameters, n = 4,304 pairs) demonstrates that $\mathcal{A}$ elevation above the class floor is a robust and broadly applicable signal of malignant transformation. The framework provides a physics-derived detection threshold ($\mathcal{A} > 1.05$) rather than a statistically trained one.

Liquid biopsy applications using cell-free DNA are a natural extension, but the input requirement must be stated precisely. Cell-free DNA in a standard blood draw is a mixture dominated (∼70%) by immune and hematopoietic cell contributions [Snyder(2016)], which dilutes any tissue-specific signal. Three input tiers are available, in decreasing order of specificity:

1. *Tissue biopsy or cytology.* Direct measurement of mean beta from the tissue of concern provides the most precise $\mathcal{A}$ computation. The G-008 validation used this input (TCGA matched tumor-normal tissue pairs).

2. *Architecture-class-specific cfDNA panel.* A targeted methylation array (e.g., 450K Illumina BeadChip) can extract the mean beta from CpG sites that are differentially

methylated in a specific architecture class relative to the immune cell background. By restricting analysis to class-specific CpG sites — identified from the reference cell database (Supplementary Table S1) — the immune dilution problem is partially circumvented and a tissue-class-specific mean beta is recovered from plasma. This is the input that would support architecture-class-specific early detection from a blood draw, replacing age-scheduled screening with physics-threshold-based triage. It requires prospective validation to characterize sensitivity and specificity at the pre-invasive stage.

3. *Bulk mean beta.* A single mean methylation beta from an unsorted blood draw provides directional information: it reflects the aggregate epigenomic state of all cfDNA contributors. At values well below the healthy reference range for all classes ($\beta \lesssim 0.68$ for most classes), the signal warrants further investigation regardless of tissue of origin. Serial bulk measurements showing a declining trend over time provide temporal evidence of accumulating epigenomic departure. This is the least specific input but requires no specialized panel.

The DCIS stratification result (low-grade $\mathcal{A} = 1.045$, high-grade $\mathcal{A} = 1.101$, threshold derived without cancer training data) illustrates the potential of tier-2 input: a secretory-class-specific cfDNA panel could in principle stratify DCIS grade from a blood draw, directing high-grade patients to treatment while sparing low-grade patients from immediate intervention. This is a specific, falsifiable prediction for prospective validation design.

Aging and longevity research. The class-specific generation rates (Table 2) constitute a physically derived prediction of tissue-specific aging rates. The terminal class drifts at 0.8% per generation; the cycling class drifts at 5.5%. This differential — not accumulated oxidative damage as a primary cause, but DNMT1 fidelity loss as the physical mechanism — is a testable mechanistic theory of tissue-specific aging. Caloric restriction, rapamycin, senolytics, and NAD+ supplementation can all be evaluated against a specific prediction: does the intervention reduce f_{C3} toward zero in the target cell class?

Pharmacological specificity. DNMT1 inhibitors (e.g., azacitidine, decitabine) are used therapeutically in myeloid malignancies. The terminal-class architecture ($n_{\text{bio}} = 24.5$) predicts that the same drug dose that shifts $\mathcal{A}$ modestly in a cycling epithelial cell will produce a substantially larger response in a neuron. This is not a clinical recommendation; it is a physically derived prediction about differential drug sensitivity that is testable in standard cell culture with Seahorse and methylation array readouts.

Regenerative medicine. The pluripotent class specification ($H_{\text{min}} = 0.9822$, differentiation dose inversion) provides a thermodynamics-based framework for optimizing iPSC reprogramming and organoid differentiation protocols. The prediction that excess transcription factor dose drives cells into aberrant states is well documented empirically; the framework provides the physical mechanism.

Developmental biology. The differentiation trajectory from pluripotent ($H_{\text{min}} = 0.9822$) to terminal ($H_{\text{min}} = 0.7728$) follows a monotone decrease in minimum entropy. This is a physically derived description of the information-theoretic cost of becoming a specific cell type. The $\mathcal{A}$ framework predicts that the C2 component grows as commitment increases and C3 closes, providing a real-time thermodynamic readout of commitment state in organoid and synthetic embryo experiments.

5.3. Limitations and experimental validation paths

The primary limitation of the current framework is the PRELIMINARY status of absolute n_{bio} values per class. The ordering is confirmed at $\rho = 0.905$ ($p = 0.002$), but absolute class-specific values require paired experiments: the same cell type measured for methylation state and metabolic flux simultaneously under controlled ATP perturbation (G-007). This is the single highest-priority experimental validation, as it tests whether the metabolic sensitivity sweep predictions are quantitatively correct in addition to structurally ordered.

The TGCT case (single non-confirmed cancer type) is not a framework failure but highlights a boundary condition: the framework applies to cells departing upward from their architecture floor. Germline-derived tumors that hypomethylate below the pluripotent floor are a structurally distinct phenomenon requiring separate treatment. This limitation is physically interpretable and does not affect the 27/28 validation result for somatic cancers.

The G-002 MCMC self-consistency validation tests internal consistency of the $H_{\min}$ derivation, not external prediction of independent data. The G-008 cancer validation constitutes the primary external test, and it is zero free parameters. The DunedinPACE fit (one free parameter, $t_{\max}$) is a structural consistency check, not an independent validation. The highest priority independent external validation is the n_{bio} absolute magnitude test (G-007), followed by prospective epigenomic fidelity index computation on TCGA cancer types not included in the G-008 database.

5.4. Relationship to existing information-theoretic biology

The present work sits in a lineage that includes Schrödinger's "What is Life?" [Schrödinger(1944)] (the founding argument that living systems maintain themselves by importing negentropy), Adami's work on information content of genetic sequences [Adami(2002)], England's derivation of thermodynamic constraints on self-replicating systems [England(2013)], and Friston's free energy principle [Friston(2010)]. The specific contribution here is the application of Landauer's principle to a concrete, measurable, genome-wide information maintenance process (DNMT1-mediated methylation copying), yielding quantitative class-specific floors that are directly testable against existing published data.

The framework does not require agreement with any of these predecessors on the deeper question of what information "is" in a biological context. It requires only the following: (1) that Landauer's principle applies to DNA methylation maintenance (a consequence of the thermodynamic reversibility argument applied to a biochemical reaction operating near equilibrium), and (2) that Shannon entropy computed from mean genome-wide methylation β is a meaningful summary statistic of epigenomic disorder. Both assumptions are independently supported by the existing literature.

6. Patent protection and data availability

The analytical methodology underlying this work — including the connection between the information actualization framework and the class-specific $H_{\min}$ registry, the n_{bio} derivation path, and the architecture class identification method — is protected under U.S. Provisional Patent Applications Serial Nos. 64/012,720 (filed March 21, 2026) and 64/014,568 (filed March 23, 2026). These provisional applications protect the methodology across 17 identified application domains and 24 architecture classes, including semiconductor, quantum computing, and genomic/cellular performance analysis.

The derived quantities reported in this paper — $H_{\min}$ values (Table 1), $\mathcal{A}$ (Eq. 4), three-component decomposition (Eqs. 7–9), cancer amplifier g_{cancer} (Eq. 10), and activation function $E(a_{\text{bio}})$ (Eq. 11) — are fully disclosed and available for research use. The E_{floor} derivation from the Landauer principle (Eq. 2) is also fully disclosed, as it follows directly from published physical constants and published DNMT1 kinetics.

The specific derivation pathway connecting E_{floor} to the class-specific $H_{\min}$ registry through the information actualization framework is available to peer reviewers under confidentiality and to qualified researchers upon written request to the author. This is consistent with standard journal practice for patent-pending methodology.

The analytical engine described in this work is released as open science for the biological domain. All MCMC scripts, the evidence database, and the analytical engine are available at `https://github.com/hmahaffeyges/IAM-Validation` (Zenodo DOI: 10.5281/zenodo.19547624).

7. Conclusion

We have derived the first first-principles thermodynamic specification of mammalian somatic cell architecture classes from the Landauer cost of DNA methylation maintenance. The framework produces eight class-specific entropy floors $H_{\min}$, validated by MCMC against 37 published reference cell measurements (all 8 parameters $\hat{R} < 1.01$) and confirmed in a zero-free-parameter forward prediction against 4,304 matched tumor–normal pairs across 27 of 28 TCGA cancer types. Three structural confirmations from neural biology — neurons defining the global floor, terminal-class cancers showing the largest thermodynamic departures, and glioblastoma's entropy non-linearity validating the Shannon functional form — constitute independent mechanistic evidence beyond the aggregate statistical result.

The biological activation function $E(a_{\text{bio}}) = \exp(1 - 1/a_{\text{bio}})$ fits published DunedinPACE age-stratified data with $\chi^2/\text{dof} = 1.14$ and recovers an actualization ceiling $t_{\max} = 120.3 \pm 7.1$ years, consistent with the Gompertz–Makeham limit.

The $H_{\min}$ specification is not a statistical model. It is a physical operating constraint that describes what a mammalian somatic cell must be in order to remain itself. Every application — cancer detection, aging research, pharmacology, regenerative medicine, developmental biology, veterinary medicine — follows from this constraint as a downstream consequence of the physics, not the other way around.

The law is the same law. The substrate changes.

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Author contributions

H.W.M. conceived the framework, performed all derivations and MCMC analyses, and wrote the manuscript.

Competing interests

H.W.M. is the named inventor on U.S. Provisional Patent Applications 64/012,720 and 64/014,568 covering aspects of the analytical methodology described in this work. The derived results and biological tools are released as open science.

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Supplementary Material

Thermodynamic Operating Constraints of Mammalian Somatic Cell Classes
Heath W. Mahaffey

S1. G-002 Reference Cell Database

Table S1 lists all 37 reference cell measurements used in the G-002 MCMC validation of $H_{\min}$ values. All beta values are from the cited primary sources; none are imputed or estimated. The MCMC scripts and provenance hashes for each entry are available at `https://github.com/hmahaffeyges/IAM-Validation`.

Table S1. G-002 MCMC reference database: 37 published cell measurements across 8 architecture classes. β: published mean genome-wide methylation. $H(\beta)$: Shannon binary entropy. All values from primary sources as cited. No values imputed.

Cell type	Class	β	$H(\beta)$	Primary source
H1 ESC	stem_pluri	0.420	0.9789	Lister et al. 2009 *Nature* [Lister(2009)]
H9 ESC	stem_pluri	0.410	0.9741	Lister et al. 2009 *Nature* [Lister(2009)]
iPSC Yamanaka P3–5	stem_pluri	0.435	0.9834	Prigione et al. 2010 [Prigione(2010)]; Lister et al. 2011 [Lister(2011)]
iPSC sendai P10	stem_pluri	0.428	0.9811	Lister et al. 2011 *Nature* [Lister(2011)]
HSC CD34$^+$ young	stem_adult	0.710	0.8680	Roadmap E035 [Roadmap(2015)]
HSC CD34$^+$ old	stem_adult	0.685	0.8936	Adelman et al. 2019 *Cell Stem Cell* [Adelman(2019)]
Neural stem cell (NSC)	stem_adult	0.720	0.8565	Zheng et al. 2016 [Zheng(2016)]; Roadmap E007 [Roadmap(2015)]
Intestinal stem LGR5$^+$	stem_adult	0.700	0.8816	Hata et al. 2020 *Nat. Genet.* [Hata(2020)]
Muscle satellite cell	stem_adult	0.715	0.8622	Bigot et al. 2015 *Cell Rep.* [Bigot(2015)]
CMP myeloid progenitor	progenitor	0.720	0.8565	Roadmap E029 [Roadmap(2015)]
GMP granulocyte prog.	progenitor	0.730	0.8415	Roadmap E030 [Roadmap(2015)]

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Supplementary Table S1 continued

Cell type	**Class**	β	$H(\beta)$	**Primary source**
Neural progenitor (NPC)	progenitor	0.715	0.8622	ENCODE [ENCODE(2012)]; Lister et al. 2013 [Lister(2013)]
Erythroid progenitor	progenitor	0.725	0.8490	Roadmap E034 [Roadmap(2015)]
Cortical neuron mature	terminal	0.780	0.7951	Kozlenkov et al. 2014 *Hum. Mol. Genet.* [Kozlenkov(2014)]
Frontal cortex neuron	terminal	0.782	0.7565	Lister et al. 2013 *Science* [Lister(2013)]
Cerebellum neuron	terminal	0.775	0.8058	Roadmap E068 [Roadmap(2015)]
Cardiomyocyte adult	terminal	0.768	0.8175	Movassagh et al. 2011 *NEJM* [Movassagh(2011)]
Skeletal muscle type I	terminal	0.760	0.8325	Roadmap E100 [Roadmap(2015)]
Colon epithelial normal	cycling	0.730	0.8415	TCGA COAD [TCGA(2012b)]; Roadmap E075 [Roadmap(2015)]
Small intestine epithelium	cycling	0.725	0.8490	Roadmap E085 [Roadmap(2015)]
Keratinocyte basal	cycling	0.720	0.8565	Roadmap E058 [Roadmap(2015)]
Bronchial epithelial	cycling	0.728	0.8445	Roadmap E096 [Roadmap(2015)]; ENCODE [ENCODE(2012)]
Colon epithelial inflamed	cycling	0.695	0.8884	Hahn et al. 2008 [Hahn(2008)]
$CD4^+$ T naive	immune	0.730	0.8415	Roadmap E043 [Roadmap(2015)]
$CD8^+$ T memory	immune	0.740	0.8265	Roadmap E048 [Roadmap(2015)]
$CD4^+$ T effector	immune	0.700	0.8816	Roadmap E044 [Roadmap(2015)]
NK cell	immune	0.735	0.8340	Roadmap E046 [Roadmap(2015)]
B cell naive	immune	0.725	0.8490	Roadmap E031 [Roadmap(2015)]
Neutrophil	immune	0.760	0.8325	Roadmap E030 [Roadmap(2015)]

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Supplementary Table S1 continued

Cell type	Class	β	$H(\beta)$	Primary source
Hepatocyte primary	secretory	0.740	0.8265	Roadmap E066 [Roadmap(2015)]
Hepatocyte NAFLD	secretory	0.710	0.8680	Ahrens et al. 2013 *Nat. Commun.* [Ahrens(2013)]
Pancreatic beta cell	secretory	0.735	0.8340	Volkmar et al. 2012 *EMBO J.* [Volkmar(2012)]
Acinar cell pancreas	secretory	0.730	0.8415	Roadmap E098 [Roadmap(2015)]
Fibroblast IMR90 P4	stromal	0.720	0.8565	Lister et al. 2009 *Nature* [Lister(2009)]
Fibroblast IMR90 P16	stromal	0.695	0.8884	Cruickshanks et al. 2013 *Nat. Genet.* [Cruickshanks(2013)]
Aortic endothelial	stromal	0.728	0.8445	Roadmap E065 [Roadmap(2015)]
Lung fibroblast normal	stromal	0.715	0.8622	Edelman 2018 [Edelman(2018)]; Roadmap E056 [Roadmap(2015)]

S2. MCMC Chain Specifications

Table S2 summarizes the MCMC configuration for all four validation chains. All chains used the *emcee* affine-invariant ensemble sampler [Foreman-Mackey(2013)]. All scripts are available at the GitHub repository with deterministic random seeds for full reproducibility.

Table S2. MCMC chain specifications for all four validation analyses.

Chain	Parameters	Walkers	Burn-in	Production	Chains	Result
G-002 (H_{min})	8 (H_{min} per class)	64	1,000	10,000	5	All $\hat{R} < 1.01$
G-008 (cancer)	0 (forward pred.)	—	—	—	—	27/28, $n = 4{,}304$
n_{bio} ordering	1 (ρ test)	—	—	—	—	$\rho = 0.905$, $p = 0.002$
$E(a_{\text{bio}})$	1 (t_{max})	32	500	10,000	5	$t_{\text{max}} = 120.3 \pm 7.1$ yr

S3. Sensitivity analysis: $\mathbf{H_{min}}$ uncertainty

To assess the robustness of the G-008 cancer detection results to uncertainty in H_{min}, we repeated the P1 direction test with H_{min} scaled by ±5% across all classes simultaneously. Results:

H_{min} scaling	P1 confirmed	Floor breach ($\mathcal{A}/\mathcal{A}_{\text{norm}} > 1.20$)
×0.95	27/28	25/28
×1.00	27/28	25/28
×1.05	27/28	25/28

The P1 direction result is completely insensitive to H_{min} uncertainty because the direction prediction compares tumor and normal tissue from the *same* H_{min} denominator; a common scale factor cancels exactly. The floor breach count is also insensitive because the relative comparison is scale-invariant. This confirms that the 27/28 result is a structural property of the entropy curve and the published beta values, not an artifact of the specific H_{min} calibration.